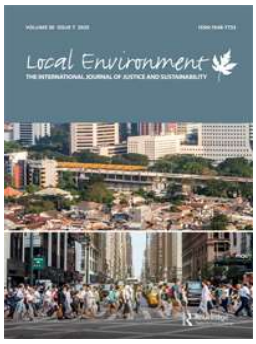

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The social perspectives of energy communities in EU policy and research – a scoping review

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ABSTRACT
The Clean Energy for All Europeans package, launched by the EU in 2019, provides a legal framework to guide the creation and management of energy communities (EC). According to the package, ECs are expected to aid a transition towards a renewable energy system by involving and empowering citizens, and thereby address issues such as poverty, social equity and "energy segregation". In technical and economic terms, on the other hand, ECs are mainly supposed to enable clusters of actors to collectively invest in technology for local renewable energy systems and share coordination benefits. Previous research indicates that ECs are not necessarily associated with the social perspectives and benefits as expected by the EU. In this scoping literature review, the aim is to investigate to what extent researchers that have studied ECs between 2015 and 2021 are highlighting the social perspectives with and benefits from ECs that are expected by the EU, as stated in the Clean Energy for All package. A total of 433 abstracts were screened and coded in accordance with a coding scheme regarding the dominant perspectives used by researchers when describing ECs and the outcomes expected from implementing them. The results show that technical perspectives dominate the research field, while social and institutional perspectives are less prominent. We conclude that social perspectives are still under-researched and somewhat neglected in research. It can thus be questioned if promotion of ECs will, by necessity, reduce energy poverty, increase social equity and hamper energy segregation.

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KEYWORDS
Energy transition; energy communities; energy justice; sustainability; social perspectives; scoping review

Highlights

- Social benefits not a necessity from promoting energy communities.
- Systematic scoping literature review on energy community research.
- Technical (followed by economic) perspectives dominate energy community research.
- Social and institutional perspectives are less prominent.

1. Introduction

The Clean Energy for all Europeans package was put forward by the European Commission in 2019 (European Commission 2019). In this package, the EU provides a policy framework for the creation

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and management of so-called energy communities (EC). In a broad sense, this EU policy is part of the overall aim to achieve a transition towards renewable and socially just energy systems. This is to be done partly by more decentralised and self-organized modes of energy production, distribution and consumption. In a more specific sense, EU member states have been given the task to implement national legislation to support the development of two forms of ECs; "citizen energy communities" and "renewable energy communities", following Directive (EU) 2019/944 and Directive (EU) 2018/2001, respectively. These directives differ in some ways, but ECs can, in both directives, be understood as a way to organise collective energy actions around open, democratic participation and governance (Bauwens et al. 2022). Furthermore, the EU-directives state that the purpose of both the above-mentioned types of ECs is to provide environmental, economic and social benefits for its members. In this review, the social benefits are in focus.

Social perspectives are, in the EU directives, considered important both for the definition of what an EC should be and for what the expected outcome is. It is clear from the EU's perspective that the social perspectives can be perceived as the fuel of the process that initiates and drives an EC. An EC cannot exist without actors that collaborate. The collaboration is important to increase the acceptance of, and support for measures in the transition towards a renewable energy system. ECs are expected to directly engage consumers and facilitate the uptake of new technologies and consumption patterns (European Parliament, Directive EU 2018/2001, paragraph 70). ECs are also, in the long run, seen by the EU as a tool to reduce energy poverty, "energy segregation", and increase social equity. EU, in particular, stresses the importance of including vulnerable and energy poor customers in ECs (Lazdins, Mutule, and Zalostiba 2021, paragraph 67).

The EU is not uniquely positioned to address the social perspectives that are important for energy system development. In fact, it has been previously acknowledged that acceptance, support, and participation of citizens are essential to manage ongoing energy transitions successfully (Bauwens, Gotchev, and Holstenkamp 2016; Kalkbrenner and Roosen 2016; Walker and Devine-Wright 2008) and that energy transition needs to be addressed within a overal socio-ecological system framework (Ostrom 2009).

Our impression is, however, that other perspectives besides social, are generally more frequently associated with EC's, including technical, economic, environmental and organisational perspectives. In technical and economic terms, ECs are generally (meaning not specifically by the EU) described as clusters of actors that collectively invest in technology for renewable energy (e.g. solar panels), who share coordination benefits of more diverse loads, load management technology, storage solutions and energy production facilities. The main purpose of ECs seems to be interpreted as providing new legal and business arrangements for community members to invest in new energy technologies, share energy and reduce operational costs. The use of new technical solutions is expected to increase renewable energy use and improve load matching.

The social perspectives have also been less frequently addressed in research. The research literature on ECs published before the launch of the Clean Energy for All Package in 2019 indicates that the description of and expected outcomes from ECs are, in general, connected to be environmental, technical and economic perspectives, rather than social. For example, Seyfang, Park, and Smith (2013) surveyed several community energy projects in the UK and concluded that economic goals (e.g. saving money on energy bills) are the most commonly cited objectives in these projects. Economically related benefits were also found to be the most prominent perspectives in reviews presented by Brummer (2018) and Berka and Creamer (2018). The latter finds that more socially oriented perspectives such as "empowerment" and "access to affordable energy" are found to be the least studied impacts from ECs (Berka and Creamer 2018). Other and more recent reviews indicate in a similar way that environmental, technical and economic perspectives in relation to ECs occur more frequently than social perspectives (Busch et al. 2021; Gjorgievski, Cundeva, and Georghiou 2021; Lazdins, Mutule, and Zalostiba 2021).

Based on the research presented above, there are reasons to question the logic presented by the EU and the assumption that ECs will result in social benefits. There is, of course, a possibility that the

EU expects that implementing the Clean Energy for All policy in itself will be a driver for achieving social benefits from ECs. Regardless, it is interesting to investigate to what extent the expectations for social benefits described in the EU-policy from 2019 have influenced the general understanding as described in literature of what ECs can achieve and how they are constituted. This is the research gap addressed in this literature review.

We argue that the research published within the scientific community is an appropriate empirical material to study in order to obtain the least biased descriptions of the understandings and expected outcomes of ECs. Even though it is a fact that researchers to some extent are biased and in cases influenced by both political and financial interests, the systematic and collegial process for publishing scientific work within the scientific community makes research publications a suitable empirical material for this scoping review.

1.1. Aim and research questions

In this scoping literature review, our aim is to investigate and nuance the approaches researchers take to ECs, by looking at what perspectives that are associated with the concept of ECs in the research literature with a focus on the years between 2015 and 2021. This review helps to shed light on the extent to which there is a coherence between the EU policy ideas considering ECs as they are stated in the Clean Energy for All Package, and current research approaches. More specifically, this study systematically, by using a multi-dimensional review structure, examines the current understandings of ECs in previous scientific literature of the social perspectives of ECs, and to what extent the expectations from the EU that ECs are to increase social justice and equity in energy supply are reflected in the research publications.

Empirically we ask the following questions about research publications:

1. To what extent are technical, economic, social and institutional perspectives included and combined in descriptions of what ECs are and seek to achieve?
2. To what extent are technical, economic, social and institutional perspectives analyzed as outcomes of ECs?
3. How do these perspectives on ECs, articulated in descriptions and outcomes, change over time?

The article is organised as follows. Section 2 gives a historical background of ECs, as described in scientific publications. Also, in Section 2, the current state of research and researchers' approaches to ECs in previous scientific reviews are presented, followed by a description of EC policy aspirations in key EU policy documents. In Section 3, the method we used to select and categorise the scientific publications is described. Results and analysis of the reviewed studies are presented in Section 4. Finally, Section 5 presents a concluding discussion of the results' implications and potential further work.

2. Background to the energy community concept and current state of research

Although new as an institutionalised policy idea within the EU, the idea of ECs as such is not new. There are descriptions in the research literature from the 1970s of citizens coming together in groups to operate energy systems, and until recently, it seems that social, institutional and economic perspectives have had a prominent position in descriptions of ECs within the scientific discourse (Langniss 1996; Walker et al. 2010). In an article from 2016, for instance, Van Der Schoor et al. (2016) argue that ECs are consumer-empowering and community-driven initiatives that can play a key role in social innovation as they reflect a fundamental shift in consumer behaviour.

During the 1970s, interest in local, small-scale and collective approaches to sustainable energy systems increased (Walker and Devine-Wright 2008). At that time, Lovins (1977) argued that the energy issue needed to be handled through what he called a soft energy path. Something he, in the same book, describes as local, community-scale energy generation. From the 1970s onwards,

descriptions in the research literature of groups of citizens coming together and operating different energy systems have become more frequent. Sometime in the late 1990s, the term "energy community" was commonly used to describe small-scale energy production, distribution or consumption in a community framework. This concept of ECs emerged in both a policy discourse and within research literature and has often been associated with renewable energy development (Walker 2008).

Roberts et al. (2019) stress that ECs can be understood as "a way to 'organise' collective energy actions around open, democratic participation and governance and the provision of benefits for the members of the local community". In line with other parts of the literature, this definition emphasises the aim of ECs to create possibilities for the members to take the role of active producers and community developers. For example, Van Der Schoor et al. (2016) argue that ECs as consumer-empowerment and community-driven initiatives can play a key role in social innovation as they reflect a fundamental shift in consumer behaviour.

Given the possibilities of such a broad definition, we may ask what more specific ambitions, values, beliefs or *perspectives* ECs are linked to and associated with in research that uses the concept. Social, institutional or economic perspectives were prominent in descriptions of ECs or other forms of small-scale, citizen-involved collective phenomena related to energy production, distribution and consumption in literature written before 2010. An example where social, institutional or economic perspectives constitute the starting point in the research related to ECs is Langniss (1996), who highlights their economic and institutional perspectives. Walker et al. (2010) further describe that ECs mark a significant departure for climate change governance towards a more cooperative, multi-actor and bottom-up distributed energy model. We also see that when Walker and Devine-Wright (2008) describe ECs, they define the concept as based on a process dimension concerning whom a project is developed and run by (i.e. who is involved and has influence) and an outcome dimension (i.e. who benefits particularly in economic or social terms). Walker and Devine-Wright (2008) point out that "these are dimensions clearly not of the technology itself, but of the particular social arrangements through which a given technology, irrespective of its scale or cost, is being implemented and made useful".

Between 2000 and 2010, the technical perspectives on ECs become more dominant in the research literature. An example is Fazeli et al. (2011), who analyze ECs from a technical and institutional perspective. Potter (2011) discusses social justice in relation to energy and takes the reasoning further by analyzing the behavioural change in relation to technology development in order to identify cost-effectiveness in the energy community's ability to produce value.

2.1. Previous literature reviews of energy communities

In this section, the general focuses and findings from previous literature reviews are presented. These reviews are selected based on the same search terms that we used for finding peer review articles but for those that are categorised as literature reviews instead. Literature published later are also included in this overview.

In general, previous reviews of ECs or related topics (i.e. community energy, P2P electricity markets, etc.) have specific research focuses, such as social innovations (Hewitt et al. 2019), public participation research (Wuebben, Romero-Luis, and Gertrudix 2020), local impacts (Berka and Creamer 2018), technical perspectives (Gjorgievski, Cundeva, and Georghiou 2021), or business models (Reis et al. 2021), or a combination of regulation and technology (Lazaroiu et al. 2025). Furthermore, the authors generally focus on a specific type of energy technology (e.g. solar energy in Lazdins, Mutule, and Zalostiba 2021) or on specific countries (e.g. the US: Klein and Coffey 2016) and the UK, Germany and the US (Brummer 2018). Research questions are defined so as to be generally answered by analysing samples of articles that omit technical-only perspectives (Brummer 2018; Busch et al. 2021; Wuebben, Romero-Luis, and Gertrudix 2020).

Different review methodologies are claimed to be used, such as a "comprehensive literature review" (Lazdins, Mutule, and Zalostiba 2021), a systematic review (Bauwens et al. 2022; Busch

et al. 2021), a systematic literature review (Brummer 2018; Wuebben, Romero-Luis, and Gertrudix 2020), literature review (Berka and Creamer 2018; Sousa et al. 2019), explorative literature review (Hewitt et al. 2019). In a few cases, a non-specific method that is just referred to as a "review" is applied (Gjorgievski, Cundeva, and Georghiou 2021; Klein and Coffey 2016; Koirala et al. 2016; Lazaroiu et al. 2025).

A number of reviews categorise the research results as, e.g. policy, economic, technical and social perspectives (Lazdins, Mutule, and Zalostiba 2021) or as social arrangements and technical designs (Gjorgievski, Cundeva, and Georghiou 2021). Seyfang, Park, and Smith (2013) present a survey of community energy projects conducted in the UK. They argue that the project objectives fall into five broad categories: economic, environmental, social, political, and infrastructural. The results from the survey suggest that economic goals (e.g. saving money on energy bills) were the most commonly cited objectives. More recent reviews indicate that in EC research, environmental, technical and economic perspectives occur more frequently than social perspectives (Berka and Creamer 2018; Gjorgievski, Cundeva, and Georghiou 2021; Lazdins, Mutule, and Zalostiba 2021). Economically related benefits were found to be the most prominent perspectives in reviews presented by Brummer (2018) and Busch et al. (2021). More socially oriented perspectives, such as "empowerment" and "access to affordable energy", are found to be the least studied impacts from ECs (Berka and Creamer 2018). Hewitt et al. (2019) state that "the social dimension of renewable energy has a large and growing literature" but list only ten publications that focus on social perspectives. A later study, not included in Hewitt et al. (2019), shows, for instance, that solar farms and power-to-gas infrastructure can increase the acceptance of local energy communities (Azarova et al. 2019) . These conclusions are, however, hard to generalise since they are not based on systematic literature searches that include a comprehensive quantitative review of articles focusing on ECs.

Some reviews report an increasing number of articles on ECs and related concepts since 2007 (Busch et al. 2021; Sousa et al. 2019), and that research on "community energy" in particular has gained momentum since 2014. Van Der Schoor and Scholtens (2019), on the other hand, note that the number of studies with a community energy focus peaks around 2014–16 and thereafter decreases (Figure 2 in European Commission 2015).

Bauwens et al. (2022) map different definitions of the term energy community, focusing mostly on the term community, and find that there is a shift over time from a notion of the term community understood as a process that emphasises participatory aspects towards a notion of a community primarily referring to a geographical place. Also, Bauwens notes a growing focus on the economic objectives of communities rather than social or political objectives. These findings, as argued by Bauwens et al., suggest a weakening of scholars' attention to notions of communities as "transformative" and a strengthening of notions that are more "instrumental".

An interesting aspect of community energy in relation to social perspectives is given in the review by Hewitt et al. (2019). This review focuses on the impact of community energy initiatives as a societal response to geopolitical crises (such as the 1970s oil crisis or the financial crisis in 2008) or as a result of governmental actions through policy (for example, the promotion of renewable energy investments). These community initiatives have historically, to a varying degree, addressed social perspectives such as self-sufficiency, democratisation of energy, and energy-price-related citizen empowerment. However, Hewitt et al. state that even though community energy initiatives are likely to continue to act as incubators for initiatives addressing virtually all aspects of energy, large multinational energy firms remain the dominant vehicle for the delivery of the energy transition. A number of challenges for energy justice, diversity, and fair distribution of benefits and burdens have been found, such as: regulatory clarity, financial support schemes, and information to overcome thresholds that limit the diversity of REC shareholders and redistribution, and not the least: the informants' understanding and focus on diversity (Standal et al. 2023). Thus, the apparent excitement in European policy circles for "community energy" seem not to extend to a democratisation of energy or genuine empowerment of citizens.

2.2. The social perspective in EU policy on energy communities

As illustrated by the brief historical overview above, the idea and practice of coming together, investing in and sharing the benefits of energy technologies have existed for decades, albeit under many different names and in many different forms. It was only recently that the EU officially recognised the benefits of collective energy actions and began the process of absorbing these ideas and transforming them into a more coherent and comprehensive policy framework. Following the 2030 climate and energy policy framework agreed upon in 2014, the European Commission presented the "energy union strategy" (European Commission COM/2015/080), which marked the beginning of a significant overhaul of EU energy policy with the aim of speeding up the transition to a low carbon economy. A year later, in 2016, the European Commission presented the "Clean Energy for All Europeans" proposal, where European consumers were the centre of attention. The proposals promised consumers a better choice of energy supply, access to reliable energy price comparison tools, the possibility to produce and sell one's own electricity, increased market transparency, opportunities for becoming more involved in the energy system and responding to price signals and measures to protect the most vulnerable consumers.

Three years later, in 2019, eight legislative acts were presented under the "Clean Energy for All Europeans" umbrella, two of which introduced the concept of ECs into EU energy policy. In the "Clean Energy for All Europeans" report published in 2019, it is evident that the European Commission sought to stress the social benefits of the clean energy transition and to promote such benefits through the new policies. It stresses that "the deep transformation of the economy needs to be managed well to avoid social and regional disparities" and that "the clean energy transition must be fair and socially acceptable to all" (European Commission 2019, 6). These social aspirations also shine through in the legislative proposals, not least where ECs are introduced.

Directive (EU) 2019/944 on common rules for the internal market for electricity explicitly stipulates that it "*aims* [among other things] *to recognise certain categories of citizen energy initiatives at the Union level as 'citizen energy communities', in order to provide them with an enabling framework, fair treatment, a level playing field and a well-defined catalogue of rights and obligations*", recognising that "*such initiatives have delivered economic, social and environmental benefits to the community that go beyond the mere benefits derived from the provision of energy services*".

Similarly, Directive (EU) 2018/2001 on the promotion of the use of energy from renewable sources recognises that "*the participation of local citizens* […] *through renewable energy communities has resulted in substantial added value in terms of local acceptance of renewable energy and access to additional private capital which results in local investment, more choice for consumers and greater participation by citizens in the energy transition*.", whereby "*measures to allow renewable energy communities to compete on an equal footing with other producers also aim to increase the participation of local citizens in renewable energy projects and therefore increase acceptance of renewable energy*."

The European Union puts a lot of focus on energy poverty and how to ensure that costumers will be able to pay their bills, and specified in 2024 that:

> Member States shall take appropriate and non-discriminatory measures to ensure that vulnerable customers and customers affected by energy poverty can access energy sharing schemes. Those measures may include financial support measures or production allocation quota. (paragraph 8, article 15a European Union, 2024)

They also state clearly that the member states are obliged – the word "shall" is used – to make shared electricity accessible to vulnerable or energy-poor customers or citizens.

The concept of "renewable ECs" differs somewhat from that of "citizen EC", for example in that it is limited to investments in renewable energy technologies, which must also be geographically close to the renewable ECs. Citizen ECs do not have those limitations, but they are limited in operating within the electricity market. Both directives, however, stipulate that an EC is a legal entity "*based on voluntary and open participation*" which "*has for its primary purpose to provide environmental,*

economic or social community benefits to its members or shareholders or to the local areas where it operates rather than to generate financial profits" (European Parliament, Directive (EU) 2019/944). In other words, both directives formulate the necessity for ECs to include the possibility for citizens to come together, invest in and share the benefits of energy technologies. The question that drives this paper is if this view is shared by research.

3. Material and methods

This study maps out general topics brought up in peer-reviewed research articles that cover the concept of ECs published after the EU absorbed it. We apply a scoping review approach (Levac, Colquhoun, and O'Brien 2010). This includes systematically mapping published literature on energy communities, focusing on how ECs are described and what outcomes are studied.

To do this, we systematically map the thematic scope of the research literature. Although the legislative proposals surrounding ECs were formally introduced in 2019 with the Clean Energy for All Europeans package, the process started earlier. In this study, we count the publishing of "The Energy Union Strategy" (European Commission, COM/2015/080) as the beginning of this process, so all studies published before 2015 have been removed from the sample.

3.1. Sampling of literature

The first step in the mapping process was a literature search in the Web of Science Core Collection (WOS) using the term "energy communit*" within the "Topic" search field (i.e. title, abstract, author keywords, and "Keywords Plus") without any search filters applied. The search was performed on 7 December 2021 and resulted in 882 documents. Documents that were not classified as "Articles" were removed (that is, conference proceedings, book reviews, editorial materials and reviews) as we wanted to limit our mapping to peer-reviewed original research articles.

In order to focus our study on papers that relate more directly to the EU policy on energy communities, we excluded other similar terms for local engagement in energy-related issues, such as "community energy" or "local energy initiatives". This way of focus is supported by an increase in the use of the term "energy community" from 2017 onwards, which can be linked to the European Commission's adoption of the Clean Energy Package in 2016, as described in (Bauwens et al. 2022). Furthermore, it has been argued that "energy communities" should not be confused with "community energy" since "energy communities" have emerged as a concept that more specifically defines the relationship of communities with their intended energy management. The latter is not the case with the more general concept of "community energy" (Gui and MacGill 2018).

As explained in the background section, we count the publishing of "The Energy Union Strategy" (European Commission, COM/2015/080) as the beginning of this process. The final sample size before the coding was 433 studies.

3.2. Literature coding procedure

The second step in the mapping process was to code the abstracts of the 433 studies. As the work was split between five coders, the procedure started with a test round where all coders coded the same abstracts within a small subsample of studies in order to agree on a common scheme and procedure. After that, all 433 abstracts were coded in accordance with the scheme below, split equally between the five coders. In case a coder was uncertain, that abstract was marked, and a second coder looked at it so that agreement could be reached on the correct coding. We concluded that abstracts, and not all articles, were sufficient to answer our research questions.

The coding procedure consisted of classifying the studies by reading the abstracts and answering the three questions in Table 1.

Table 1. The questions and answer alternatives used in the abstract coding procedure.

Questions	Answer alternatives (codes)
1. Are ECs a central focus of the study?	Yes; No (single choice)
2. How are ECs described in terms of what they are and/or seek to achieve?	Technical; Economic; Social; Institutional (multiple choice)
3. What are the EC outcomes of interest under study?	Technical; Economic; Social; Institutional

Initially, a selection criterion was used to further refine the sample, focusing on studies where the concept of ECs was actually central to the study and not just mentioned. In other words, studies referring to "energy communities" in a wider meaning, such as a European energy community or an organisation like the European Atomic Energy Community (EURATOM), were removed. Given that we did not read the full papers, our selection criterion was quite generous. This selection process resulted in a final sample of 243 studies eligible for further classification by answering questions 2 and 3.

The analysis was made based on the delineation of four possible "analytical" or "ideal-type" *perspectives* on ECs, summarising four different frames of references in relation to what ECs are about in terms of defined problems and suggested solutions. The *technical perspective is defined by its understanding of ECs as a technical phenomenon where* technical specifications, product locations and interactions with other technical devices in energy systems are in focus as solutions. From an *economic* perspective, ECs are approached from in terms of cost and benefit and as a strategy to generate income or save costs for individuals, groups, organisations or society at large. The *social perspective* understands ECs as a social arena where people interact, and community relationships evolve as a possible solution to increasing inclusion and combating exclusion. Finally, the *institutional perspective* approaches ECs as *organised* entities, focusing on internal governance structures as well as external relations with national institutions, policies and other external stakeholders.

In order to answer the overall aim, we not only distinguish between four possible analytical perspectives on ECs, we also distinguish between two dimensions in order to study to what extent different perspectives characterise individual research publications. In Table 2 below, we therefore distinguish between technical, economic, social and institutional orientations along two dimensions – objectives and outcomes – in our analytical framework for coding and analysing research on ECs.[1] The first dimension, which corresponds to the first research question, refers to how ECs were described in individual research publications in terms of what they are and/or seek to achieve. The second dimension, which corresponds to the second research question, refers to the specific EC outcomes of interest in an individual research publication.

Accordingly, our analytical grid consists of four possible perspectives and two dimensions, as illustrated and summed up in Table 2 below. In our analysis and coding, however, each dimension could be characterised by using all four perspectives. Each individual research publication could

Table 2. Definition of codes used to classify the selected articles.

Code	RQ1) How are ECs described in terms of what they are and seek to achieve?	RQ2) What are the EC outcomes of interest under study?
Technical	As solutions to technical challenges, such as the use of locally produced electricity, etc.	Technical outcomes such as increased energy savings, supply/demand balances, power quality, battery utilisation rates, etc.
Economic	As solutions to economic challenges, such as a means to reduce costs of energy technology investments, etc.	Economic outcomes such as investment costs, payback times, electricity prices, etc.
Social	As solutions to social challenges, such as a means to empower citizens, reduce energy poverty and/or inequalities, etc.	Social outcomes, such as consumer behaviour and practices, attitudes, norms, participation, energy poverty, inequalities, etc.
Institutional	As solutions to governance challenges, such as a means to realise policies and legislative goals, improve market functionality, etc.	Institutional outcomes, such as stable, transparent, legitimate, internal and external governance structures, etc.

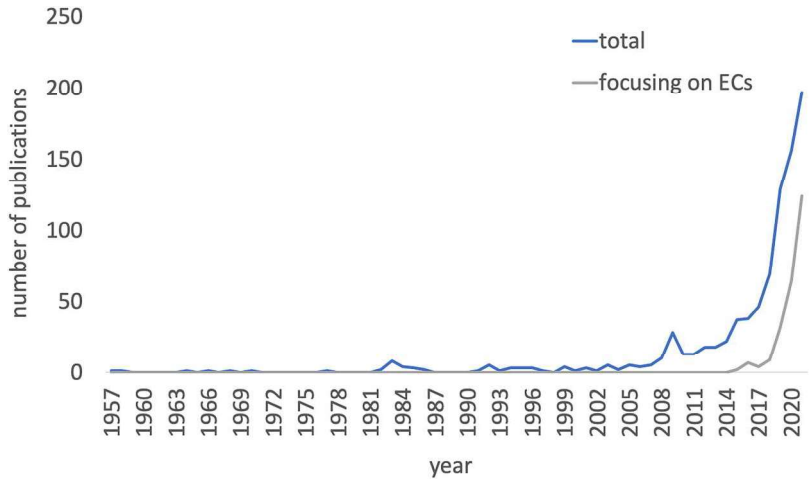

Figure 1. Number of publications per year (1957–2021) included in our sample (publications in 2022 omitted from this figure). The blue line represents all search results (n = 882) and the grey line is the number of papers focusing on ECs as defined by our selection criteria (n = 243).

consequently be more or less "pure" in terms of its EC perspective. For instance, a *purely technical approach in a specific piece of research stresses only the* technical objectives of ECs and focuses only on technical outcomes. At the other end of the spectrum, we may theoretically find publications that stress equally all four perspectives on both dimensions.

Figure 1 provides an overview of the sample of selected articles before and after removing articles that did not meet the selection criteria, ordered by year of publishing, to illustrate how the coverage of ECs in the academic literature has changed over time.

4. Results

The results from the coding of abstracts are presented in this section. Table 3 shows the share of abstracts with an EC focus that are deemed to address either single or multiple perspectives of ECs. Out of the included abstracts, 56% were deemed to focus on ECs. Out of these, 54% describe ECs with a single characteristic perspective (i.e. technical, social, economic, or institutional). The corresponding share for the expected outcome of ECs is 52%. Otherwise, the description and expected outcome are described using multiple characteristics.

Figure 2 shows the number of abstracts with an EC focus that addresses the different characteristic perspectives that were coded for. Note that the blue and orange bars, respectively, sum up to more than 243. This is because the abstracts were, in some cases, coded to address more than one dimension, both for the description of ECs and expected EC outcomes. The results presented in Figure 2 show that technical descriptions of ECs and expected outcomes are dominating. In 177 abstracts, the EC is described, completely or partially, as a technical system. The corresponding numbers for socially, economically, and institutionally characterised descriptions of ECs are 53, 65 and 81, respectively. For expected EC outcomes, the pattern is similar but less pronounced. The expected outcome of implementing ECs is

Table 3. Percentage of abstracts coded with single and multiple dimensions.

Abstracts focusing on EC's	Percentage of abstracts focusing on EC's
Addressing a single dimension in description	54
Addressing a single dimension in expected outcome	52
Addressing multiple perspectives in description	43
Addressing multiple perspectives in expected outcome	47

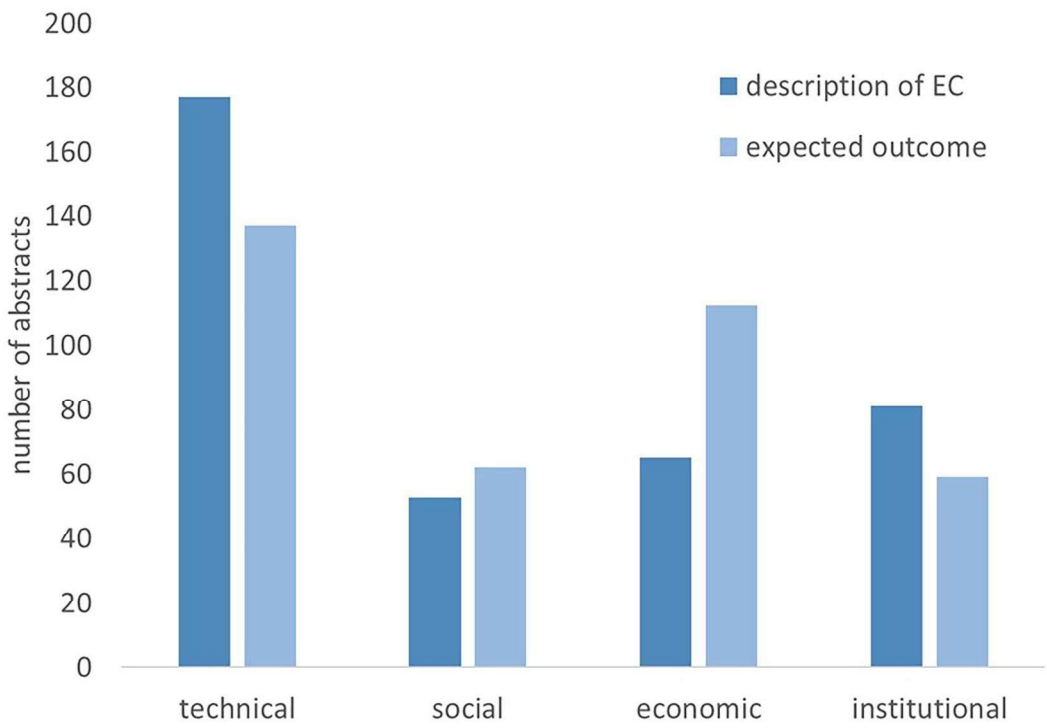

Figure 2. Numbers of scientific papers addressing different perspectives of and expected outcomes from ECs.

technical in 137 abstracts and social in 62 abstracts. Expectations of economic outcomes from ECs are, however, significantly more frequent than economic descriptions of ECs (112). The institutional perspective of expected EC outcomes is represented in 59 abstracts.

Figure 3 shows the number of abstracts coded for single and multiple combinations of perspectives for EC description and expected EC outcome, respectively. The abstracts with a technical description are significantly in the majority among the single-dimension coded abstracts. The technical dimension is also prominent for expected EC outcomes, although less prominent than for the descriptions. For the multiple-dimension coded abstracts, the technical-economic perspective is the most common, both for EC description and for expected EC outcome. Note that the number of abstracts coded with a combination of technical and social perspectives (denoted SE in Figure 3) is small. This is interesting since social and economic perspectives, in a sense, can be expected to be closely connected to each other.

In Figure 4a, the number of different characteristics represented in the abstracts' descriptions of ECs is shown and ordered by year of publication. The corresponding numbers for the expected outcomes are presented in Figure 4b. It is clear that the total number of publications that consider ECs increases from 2019 onwards. It is also clear that the technical perspectives in both EC descriptions and expected outcomes increase relative to the other perspectives from 2019. There is one exception, however. The economic dimension in the expected outcomes (Figure 4b) also increases substantially from 2019. The overall trend shown in Figure 4a and b is that technical descriptions of ECs and technical-economic expected outcomes from ECs have become much more prevalent in scientific papers published in 2019 onwards. Social and economic/institutional perspectives have also increased but from a much lower number. The dominance of technological perspectives continues to be strong. Note that since the number of publications prior to 2019 is small, the focus on different perspectives within EC research before 2019 is difficult to assess from the results presented here.

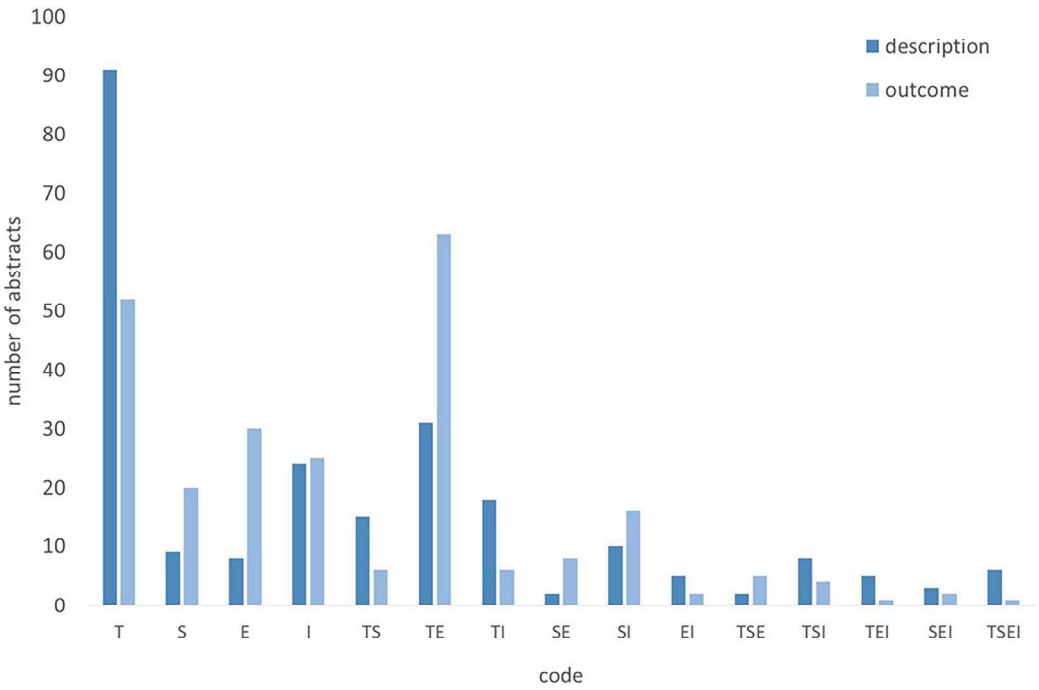

Figure 3. Number of abstracts with different codes and combinations of codes (T = technical, S = social, E = economic, and I = institutional).

As illustrated by the figures, relatively few original research articles address the social perspectives of ECs. The studies that do so present participants in energy communities as active agents who create and share important knowledge (Hyysalo, Juntunen, and Martiskainen 2018; Nielsen, Baer, and Lindkvist 2019), focus on innovations (Nielsen, Baer, and Lindkvist 2019), or on engagement, empowerment and environmental awareness (Kloppenburg and Boekelo 2019; Koirala et al. 2018).

When combining codes for definition and outcome, we obtain $12 \times 12 = 144$ possible combinations. The ten most common ones are listed in Table 4 (a table with all 144 possible combinations is available from the authors on request).

Here, we can observe that a purely technical approach to ECs is by far the most common in the research. A total of 43 articles include only technical dimensions in the definition and outcome.

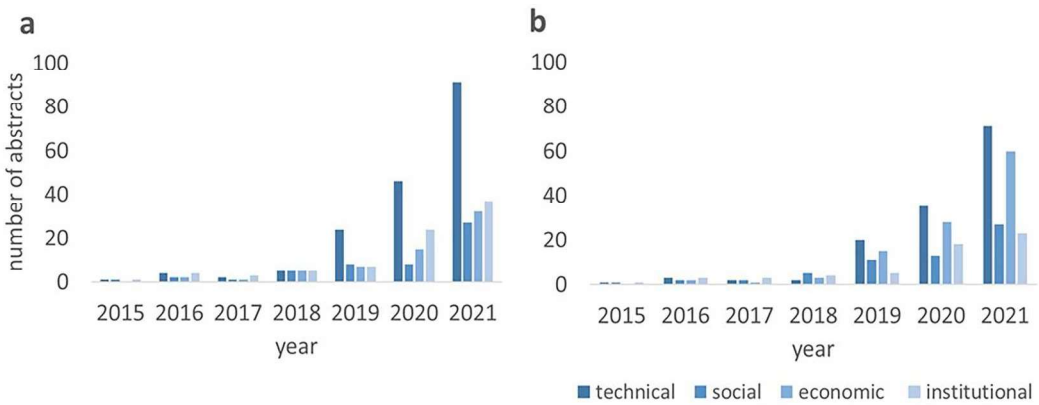

Figure 4. Number of scientific papers with different descriptions of ECs per year (a), and with different expected outcomes (b).

Table 4. The most common combinations of definitions and outcomes.

Combination	Number
Purely technical (only technical definition and only technical outcome)	43
Technical definition and technical-economic outcome	30
Technical-economic definition and technical-economic outcome	21
Purely institutional	15
Purely social	8
Purely economic	7
Technical definition and technical-economic outcome	6
Technical-economic definition and economic outcome	6
Technical-institutional definition and technical institutional outcome	4
Technical-social definition and social outcome	4

Articles that include a social dimension in the definition or outcome are more scarce and are often combined with other dimensions. However, there are eight articles that have a purely social focus, i.e. they define ECs as well as their expected outcome in social terms. Otherwise, social dimensions are often combined with technical dimensions.

This article aims to answer the following three research questions:

1. To what extent are technical, economic, social and institutional perspectives included and combined in descriptions of what ECs are and seek to achieve?
2. To what extent are technical, economic, social and institutional perspectives analysed and combined as outcomes of ECs?
3. How do these descriptions and outcomes change over time?

Regarding the first question, the results of this study indicate that there are significantly more ECs described completely or partially as a technical system than there are ECs characterised as socially, economically, and institutional communities. After the technical dimension, the institutional perspective is the second most common. Followed by the economic perspective and, lastly, the social perspective. Among articles coded with multiple perspectives, the technical-economic perspective is the most common.

The analysis of the empirical data also provides answers to the second research question, indicating that the most common expected outcome of implementing ECs is technical. The second-most expected outcome of ECs is financial, and, to the least extent, significantly lower than for technical and financial outcomes; the institutional and social dimensions occur to about mutually the same level. This pattern is also reinforced by the part of the analysis that results in multiple-dimension coded abstracts. There, the technical-economic perspective is the most common expected EC outcome.

When we look at expected results linked to institutional or social perspectives, we see that these perspectives occur to an approximately equal degree,

Regarding the third research question, our results indicate that there is a clear dividing line between articles published before or after the turn of the year 2018/2019 with regard to how much the technical perspective dominates the research articles, both regarding how they are described and regarding expected outcomes. Up to and including the year 2018, all four perspectives are equally common in research articles, but from the year 2019 onwards, the technical perspective is dominant. In addition, publications from 2019 onwards describe expectations of an economic outcome from ECs to a significantly higher degree than before 2019.

5. Concluding discussion

In this study, we have examined the perspectives of ECs that have been given attention and have been combined in the research literature and close in time to the launch of EU's Clean Energy for All Package in 2019. The abstracts of scientific articles published between 2015 and 2021 has

been reviewed and analysed to investigate the extent to which technical, economic, social and institutional perspectives are considered and combined in the understanding of ECs.

The overall conclusion of our analysis is that technical perspectives dominate over social, institutional and economic perspectives in the research literature from 2015-2021, both in terms of how ECs are described and the expected outcomes of ECs. Technical and economic perspectives dominate the analysis of the combination of perspectives in the research literature. Despite the observed emphasis on social perspectives in EU policy from 2015 onwards, the technical perspective has increased its relative dominance over the years. If there is a the link between what policymakers and practitioners do and what researchers focus on as a result of funding opportunities and empirical cases to study, it is also possible that the social perspectives of ECs have been lost in translation from policy to practice. We interpret however the result of our study as an indication of a sceptical approach to the importance of social perspectives of ECs within the scientific community.

Our conclusions with regard to the research questions posed in our aim and hypothesis section are as follows:

- Technical, economic, social and institutional perspectives are all represented both individually and in different combinations in the understanding of ECs within the research community.
- Social, institutional and economic perspectives are significantly less represented than technical perspectives in descriptions of ECs.
- For expected outcomes, technical perspectives are less dominant, while economic perspectives are relatively well represented.
- Economic outcomes are significantly more represented in combination with technical descriptions of ECs.

In general, the results presented here show that social perspectives of ECs have been overshadowed by an increasing focus on technical perspectives and that technical perspectives have become relatively more prominent in the research literature, especially in publications from 2019 onwards. The second most frequent perspective the articles address is the institutional dimension. Regarding the expected outcome of ECs, the second most frequent characteristic was the economic dimension. This is likely associated with the fact that the economic perspective is a characteristic strongly related to the possibility of investing in and motivating the implementation of energy technologies. The economic perspective is also strongly related to some of the social perspectives, such as energy poverty and social inequalities. Therefore, it is noteworthy, and somewhat surprising, that the combination of social and economic perspectives for expected EC outcomes is relatively scarce. This could perhaps be explained by unspoken expectations of positive social effects from positive economic effects. In other words, the expected positive economic benefits for EC actors might also indirectly reduce energy poverty and economic inequalities.

However, it is also important to relate the conclusions to more general patterns in research and science. The number of studies that consider social perspectives can be regarded as low when related to the higher quantitative amount of studies that focus virtually exclusively on the technical perspectives of ECs. However, it is relevant to point out that publications focusing on technical perspectives in general are more numerous than publications on social perspectives. The relatively low proportion of studies that consider social perspectives in ECs, compared to other perspectives, may be explained in different ways: lack of social perspectives in studies of technical solutions in general, quantitative dominance of technical studies among publications in general, that technical studies more rapidly can adjust to relevant new policies, or that social perspectives are complex and more time-consuming to study, leading to fewer studies on social perspectives in ECs and more time elapsing before formulation of the policy and study publication. The low proportion of articles that discuss social perspectives in ECs may also be a reflection of policy and practice channelled through research funding, education and industrial partnerships.

We argue that the vision presented in The Clean Energy for All Package that a large-scale implementation of ECs will achieve social benefits, needs to be critically scrutinised. Our results show that the expectations as stated in the EU's policy on ECs in the late 2010s are not mirrored in research litterature. We further argue that more research on how social benefits are to be achieved by the implementation of EC's is required. This to identify essential aspects and alternative support for a socially accepted technological-economic approach to the necessary transformation of energy production and consumption. We, therefore, call for greater awareness of social perspectives in studies on technical, economic, and institutional aspects.

Note

1. In their analysis of how ECs are conceptualised in research, Bowens et al [1] distinguish in a similar way between how researchers conceptualise meanings, activities and objectives of communities.

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